OmniRAG Enterprise AI Platform v2.0 - Debug & Setup Guide

Quick Start (Final Working Commands)

```
cd capstone-enterprise-rag/capstone-enterprise-rag
docker compose build --no-cache
docker compose up -d --force-recreate
docker exec -it pg_pinecone_local psql -U postgres -d pinecone_db -c "SELECT
# Must return 14

curl http://localhost:8080/api/health
# {"service":"enterprise-rag","version":"2.0","status":"UP"}

curl -X POST http://localhost:8080/api/ask -H "Content-Type: application/jso
# Returns: "20th Oct: Acme Corp complained about pricing..." latency 1014ms
```

7 Bugs Fixed (See Word doc for details)

1. **Bean Override**: allow-bean-definition-overriding true, delete VectorStoreConfig, use table-name enterprise_docs
2. **Wrong DB**: Use pinecone_db not postgres DB, never down -v
3. **VectorStore Table**: application.yml table-name: enterprise_docs, schema-name: public
4. **Filter Bug**: Use FilterExpressionBuilder.eq("source", table).build() not string concat - fixed 206ms Not Found -> 1s correct
5. **Frontend 404**: Dockerfile must COPY nginx.conf, then up -d --force-recreate
6. **HTML not JSON**: nginx.conf proxy_pass http://backend:8080 (no trailing /api/)
7. **Latency**: topK 4, threshold 0.0, warmup Ollama

Architecture

React (5173) --nginx proxy--> Spring Boot (8080) --PgVectorStore--> postgres:5433/pinecone_db.enterprise_docs (14 docs, HNSW) + Ollama

llama3.2:3b

Sources: crm_docs (3), odoo_docs (3), parking_docs (3), medical_kb (3), whatsapp_docs (2), ncert_docs (1)

Screenshots

See Word document for full screenshots:

- Fig1: Initial UI with 6 sources, analytics 92% hit rate
- Fig2: Ask All Sources returning CRM demo

Capstone Demo Queries

Use these 5 for video:

1. crm_docs: Acme pricing issue
2. odoo_docs: Odoo inventory issue
3. parking_docs: parking slot allocation
4. medical_kb: diabetes treatment
5. ask-all: Acme pricing issue

Frontend: http://localhost:5173
Backend: http://localhost:8080/api/health